Fault Diagnosis of Wheeled Robot Based on Prior Knowledge and Spatial-Temporal Difference Graph Convolutional Network

Zhaoming Miao, Yingxiang Xia, Fengyu Zhou, and Xianfeng Yuan

***Abstract*—The critical issue of wheeled robot fault diagnosis is to comprehensively evaluate its health condition using multisensor data, but traditional deep learning-based methods are hard to model the relationships among sensor measurements. Unlike these methods, the graph convolutional network (GCN), which uses the graph-structured data along with the association graph as input, is more efficient for relationship modeling. However, existing GCN-based fault diagnosis methods suffer from the following weaknesses: the association graphs are obtained according to the similarity of data samples or their features, which cannot guarantee accuracy; and these models are focused on spatial correlations and neglect temporal correlations. To address these problems, we propose to construct the association graph based on prior knowledge, *i.e.*, a simplified mathematical model of the wheeled robot. Moreover, we develop a spatial-temporal difference graph convolutional network (STDGCN) for wheeled robot fault diagnosis. This network contains a difference layer that utilizes localized difference properties for feature enhancement, and the spatial-temporal graph convolutional modules are introduced to jointly capture the spatial-temporal correlations. To verify the effectiveness of the STDGCN for fault diagnosis, experiments are carried out, and the results show that the STDGCN achieves superior performance.**

***Index Terms*—Deep learning, graph convolutional network, intelligent fault diagnosis, wheeled robot.**

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I. Introduction

With the rapid development of robot technology, various wheeled robots are applied to a wider range of fields, including home services, industry, and aerospace. However, like other electrical systems, wheeled robots are prone to faults with the service time growing. These faults could subsequently affect the robot's performance and even cause serious incidents. Therefore, it is worthwhile to investigate the fault diagnosis method of wheeled robots.

Recent advances in data acquisition hardware, neural network theory, and parallel computation technology have empowered data-driven fault diagnosis technologies to achieve great breakthroughs [1], especially for deep learning (DL)-based methods [2], [3], [4]. Many DL-based methods have found eminent applications in various fault diagnosis cases, such as gearboxes [5], [6], [7], bearings [8], [9], transformer [10], etc. For example, Wang et al. [11] developed a deep neural network-based framework to monitor the conditions of wind turbine gearboxes. Shao et al. [12] proposed a novel convolutional deep belief network for bearing fault diagnosis. Based on 1-D convolutional neural network, Huang et al. [13] introduced a fault diagnosis method to identify the fault states of high-speed train bogie. Peng et al. [14] proposed a multibranch and multiscale convolutional neural network for fault diagnosis of wheelset bearings.

In most fault diagnosis cases mentioned previously, only using a single type of sensor signals, such as vibration signals or current signals, can achieve satisfying diagnosis performances. Unlike these studies, in our case, it is difficult to accurately classify the faults of a wheeled robot based on single type of sensor signals, because the wheeled robot is a highly coupled electromechanical system with nonlinearity and time-varying properties. Since the wheeled robot usually equipped with a variety of heterogeneous sensors, it is necessary to adopt multisensor data to comprehensively evaluate its health condition. However, the discriminative feature extraction and fusion of multisensor data is a difficult problem, and it is hard to model the relationships of sensor signals via traditional methods.

Fortunately, the graph convolutional network (GCN) could solve this problem. The GCN was developed by Bruna et al. [15] and Defferrard et al. [16]. By introducing the association graph, the GCN could aggregate the neighborhood node features of the central node to generate new node features, and the correlations among nodes could be extracted for feature representation

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during this process [17]. Currently, the GCN has been successfully used in applications, such as traffic prediction [18] and knowledge graph embedding [19]. Inspired by these achievements, several recent studies have attempted to apply the GCN for fault diagnosis and remaining useful life prediction. Yu et al. [20] proposed a fast deep GCN to deliver fault diagnosis of wind turbine gearbox. The proposed method decomposed the original vibration signals using the wavelet packet, and then, introduced the fast deep GCN to extract the features. Liao et al. [21] presented a GCN-based method for transformer fault diagnosis, in which the adjacency matrix was utilized to represent the similarity metrics between unknown samples and labeled samples. Zhang et al. [22] developed a deep GCN for fault diagnosis of roller bearings, where the collected acoustic signals were transformed into graphs and the edge weights represented the similarity between connected vertices. In [23], multisensor data were used to design multiple k-nearest neighbor graphs, and an improved multichannel graph convolutional network was proposed to extract graph features.

The aforementioned GCN-based approaches determine the association graphs based on the similarity of data samples or their features. However, the association graphs obtained in this way cannot accurately reflect the actual relationships, which would result in negative effects on the network training and even limit the diagnosis performance. In addition, most of these methods are inclined to mine spatial correlations, while ignoring temporal correlations that are also essential for fault classification.

To settle the aforementioned problems, we study a new strategy to design the association graph and devise a spatial-temporal difference graph convolutional network (STDGCN) for wheeled robot fault diagnosis. First, we build a simplified mathematical model of the wheeled robot and determine the robot data association graph based on the model. Then, we convert the wheeled robot sensor measurements into spatial-temporal graphs, and a difference layer is introduced into the STDGCN for feature enhancement of graph nodes by calculating the backward difference features. Moreover, the spatial-temporal graph convolutional module (STGCM) is developed to integrate fault information along the spatial and temporal dimensions. Finally, fault diagnosis is realized by the proposed STDGCN. Comparative experiments between STDGCN and several existing fault diagnosis methods are conducted. Experimental results show the superiority of the STDGCN. The main contributions of this article are summarized as follows.

1) A novel association graph construction method is proposed, which uses the prior knowledge of the wheeled robot and enables the GCN-based method to explicitly exploit the spatial correlations among sensor signals.
2) A difference layer that takes advantage of localized difference properties is designed for feature enhancement of graph nodes, which calculates multiorder backward difference features.
3) The spatial-temporal graph convolutional module is introduced to boost mode performance by capturing the spatial-temporal correlations jointly.

The remainder of this article is arranged as follows. Section II introduces the basic theory of the GCN. Section III describes the proposed association graph construction method and the developed STDGCN model in detail. After that, the effectiveness and superiority of our proposal are validated by ablation studies and comparative experiments in Section IV. Finally, Section V concludes this article.

II. Graph Convolutional Network (GCN)

Similar to the convolutional neural network (CNN), the essential goal of the GCN is to extract feature information from graph data through convolution operation. The difference lies in that GCN deals with the non-Euclidean data. Due to the lack of translation invariability in graph data, it is difficult to directly define the convolution operation through spatial domain. Therefore, researchers define the convolution operation on graph data in the spectral domain.

Consider the convolution operation $f * x$ on the spatial domain, where f represents the convolution kernel and x represents the input data. Since the Fourier transform of the graph signals, which have been processed by the convolution operation, is equal to the multiplication of these signals that transformed into the spectral domain, the convolution operation $f * x$ in the spatial domain can be expressed as

$$\begin{aligned} f * x &= F^{-1}(F(f) \odot F(x)) \\ &= U\left(\left(U^T f\right) \odot \left(U^T x\right)\right) \end{aligned} \tag{1}$$

where F and F^{-1} represent the Fourier transform and the inverse Fourier transform, respectively, U denotes the Fourier basis, and $\odot$ represents the element-wise multiplication operation. In order to introduce the association graph into the neural network, the Fourier basis used by the GCN is obtained by the Laplacian matrix of the association graph, and its specific calculation method is as follows: suppose $L = D - A$ is the Laplacian matrix of a graph, it can be normalized as $L = I_N - D^{1/2} A D^{1/2} \in \mathbb{R}^{N \times N}$, where I_N represents an unit matrix, A denotes the adjacent matrix, and D is the degree matrix, $D_{ii} \in \sum_j A_{ij}$. Then, the Fourier basis U along with the eigenvalue matrix Λ can be obtained by the eigenvalue decomposition

$$U \Lambda U^T = L, \Lambda = \operatorname{diag}\left([\lambda_0, \ldots, \lambda_{N-1}]\right). \tag{2}$$

According to the properties of the Laplace matrix, U is a set of orthogonal matrices that satisfies the basic mathematical requirements of Fourier transform. Let $g_\theta = \operatorname{diag}(U^T f)$, (1) can be simplified to

$$f * x = U\left(\left(U^T f\right) \odot \left(U^T x\right)\right) = U g_\theta U^T x. \tag{3}$$

In the process of graph convolution, the eigenvalue decomposition of the Laplace matrix is an important step, but when the graph scale is large, the cost of eigenvalue decomposition of the Laplace matrix is very expensive. To address this problem, Defferrard et al. [16] proposed to approximate g_θ using Chebyshev polynomials T_k, thus (3) can be alternatively expressed as

$$f * x = U g_\theta U^T x \approx \sum_{k=0}^{K} \theta_k T_k\left(U \hat{\Lambda} U^T\right) x = \sum_{k=0}^{K} \theta_k T_k\left(\hat{L}\right) x \tag{4}$$

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where θ is the polynomial coefficient. $\hat{L} = 2\,L/\lambda_{\max} - I_N$ and $\lambda_{\max}$ denotes the Laplacian matrix's maximum eigenvalue. The Chebyshev polynomials can be defined as $T_K(x) = 2xT_{K-1}(x) - T_{K-2}(x)$, $T_0(x) = 1$, and $T_1(x) = x$. Using the Chebyshev polynomial approximate expansion solution is equivalent to using convolution kernel to extract the order 0 to K neighbor information around the center of each node in the graph.

Kipf and Welling [24] proposed to further simplify Chebyshev polynomials by setting $\lambda_{\max} = 2$ and $k = 1$, which means aggregating the first-order neighborhood node feature of central nodes. In this way, (4) can be simplified to

$$\begin{aligned} f * x &\approx \theta_0 x + \theta_1 \left(2\,L/\lambda_{\max} - I_N\right) x \\ &\approx \theta_0 x - \theta_1 \left(D^{-\frac{1}{2}} W D^{-\frac{1}{2}}\right) x. \end{aligned} \tag{5}$$

Through setting the parameter $\theta = \theta_0 = -\theta_1$, a single parameter formula $f * x \approx \theta_1 (I_N + D^{-\frac{1}{2}} W D^{-\frac{1}{2}}) x$ could be obtained. In order to facilitate the training of network during backward propagation, the parameters W and D are usually renormalized by $\tilde{W} = W + I_N$ and $\tilde{D}_{ii} = \sum_j \tilde{W}_{ij}$, respectively. Finally, the convolution operation $f * x$ in the GCN can be mathematically defined as

$$f * x \approx \theta \left(I_N + D^{-\frac{1}{2}} W D^{-\frac{1}{2}}\right) x = \theta \left(\tilde{D}^{-\frac{1}{2}} \tilde{W} \tilde{D}^{-\frac{1}{2}}\right) x. \tag{6}$$

III. Proposed Fault Diagnosis Method

Wheeled robot sensor measurements contain rich relation information, and our interest is to model the spatial-temporal dependencies from different sensor signals. Wheeled robot sensor measurements can be defined as spatial-temporal graphs, and this process contains two steps. First, the sensor measurements within a time slice are connected with edges according to the relationship of corresponding sensor variables. Then, each sensor measurement is connected to the same sensor measurements in the consecutive time slices. Here, we begin with the robot data association graph construction, that is, modeling the relationships of sensor variables, and then, introduce the proposed STDGCN for wheeled robot fault diagnosis in detail.

A. Association Graph Construction

The fundamental of using GCN-based methods is to obtain the association graph. In this article, we propose to build the association graph according to prior knowledge, i.e., the mathematical model of wheeled robot. This article takes a Mecanum wheeled mobile robot, which is widely applied in industry and service fields, as the experimental platform. Those sensor data that contain the robot running state information are chosen for fault diagnosis. The adopted sensor variables are listed in Table I, where the subscript numbers in variables indicate the motor label of the wheeled robot, and the subscript letters denote the coordinate axes.

We now advance to the construction of mathematical model. Our purpose is to use the mathematical model of the wheeled robot to reflect the relationships among sensor variables. Therefore, a simplified mathematical model, which ignores the impact of environmental factors, can satisfy our needs.

TABLE I
Adopted Robot Sensor Variables

Physical meanings of variables	Variable			
Encoder pulse	E_1	E_2	E_3	E_4
Wheel velocity	v_1	v_2	v_3	v_4
Motor torque	T_1	T_2	T_3	T_4
Robot acceleration	a_x	a_y	a_z	-
Robot angular velocity	w_x	w_y	w_z	-

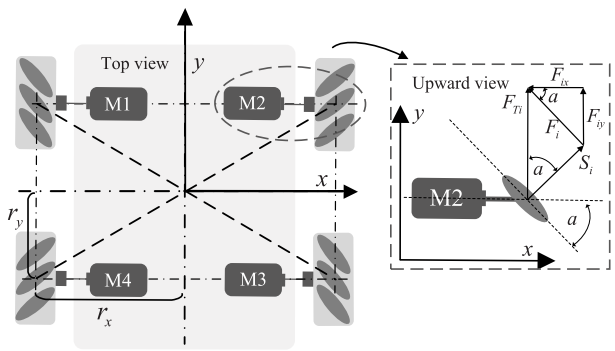

Fig. 1. Coordinate of the Mecanum wheeled mobile robot and view of forces acting on the wheel roller that in contact with the ground.

Making the assumption that the wheeled robot is rigid and all points on the robot rotate around the instantaneous center of rotation [25]. Ignoring the influence of friction. Let (x, y, z) be the coordinate axis of the Mecanum wheeled mobile robot with the geometric center as the origin. Fig. 1 presents the top view of the Mecanum wheeled mobile robot and an upward view of the motor driving force F_{Ti} acting on the wheel roller, where F_{Ti} can be divided into the roller effective drive force F_i and the ineffective slip force S_i. F_{ix} and F_{iy} are the drive force in the directions x and y, respectively.

According to Fig. 1, the mathematical model, including the dynamic and kinematic model of the wheeled robot, is built as shown in (7)–(15), and the meanings of undefined variables and functions in these equations are presented in Table II. Note that the modeling process is not the focus of this article, and some detailed modeling processes can be found in [25].

$$\begin{cases} F_x = \sum_{i=1}^{4} (-1)^{i+1} \mathrm{sig}(\theta_i) F_{Ti} \sin a \cos a \\ F_y = \sum_{i=1}^{4} \mathrm{sig}(\theta_i) F_{Ti} (\sin a)^2 \end{cases} \tag{7}$$

$$F_{Ti} = T_i r_{\mathrm{motor}}, i = 1, 2, 3, 4 \tag{8}$$

$$a_j = F_j / M, j = x, y \tag{9}$$

$$a_{\omega z} = I \left(\sum_{i=1}^{4} \mathrm{sig}(\theta_i) F_{Ti} \sin a - F_x - F_y \right) \tag{10}$$

$$\omega_z = \sum_{n=0}^{N} K a_{\omega z}^{n} \Delta t \tag{11}$$

$$v_i = C_i \frac{\Delta E_i}{\Delta t}, i = 1, 2, 3, 4 \tag{12}$$

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TABLE II
MEANING OF VARIABLES AND FUNCTIONS IN THE EQUATIONS

Variable	Meaning
a	The angle between the wheel roller and the rotation axis of the roller
F_{Ti}	The force on the wheel roller due to the i-th motor torque T_i
$\mathrm{sig}(\cdot)$	A sign function denotes the wheel rotational direction and determined by the right-hand rule
M	The total mass of the Mecanum wheeled mobile robot
θ_i	The angular rotation made by wheel i
r_{motor}	The radius of the motor
$a_{\omega z}$	The robot angular acceleration in the direction z
N	The sampling times
I	A constant dependent on the moment of inertia and the robot shape
K	A constant dependent on the robot shape and the mass distribution
Δt	The time interval between sensor sampling
C_i	A constant of the i-th encoder depends on the wheel radius and the number of pulses per encoder revolution
ΔE_i	The change in the number of encoder pulses in a time interval of the ith encoder

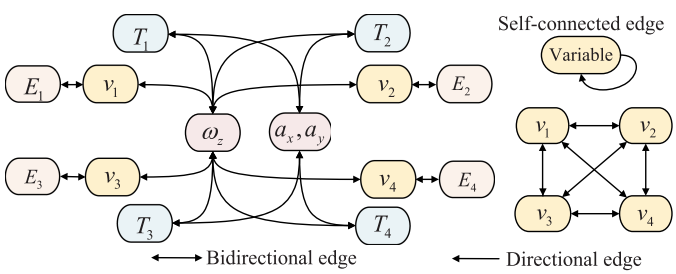

Fig. 2. Robot data association graph.

$$\begin{bmatrix} v_1 & v_2 & v_3 & v_4 \end{bmatrix}^{\mathrm{T}} = R \begin{bmatrix} v_y & v_x & \omega_z \end{bmatrix}^{\mathrm{T}} \tag{13}$$

$$\begin{bmatrix} v_y & v_x & \omega_z \end{bmatrix}^{\mathrm{T}} = \frac{1}{4} R^{\mathrm{T}} \begin{bmatrix} v_1 & v_2 & v_3 & v_4 \end{bmatrix}^{\mathrm{T}} \tag{14}$$

$$R = \begin{bmatrix} 1 & \cot a & l_0 \\ 1 & -\cot a & -l_0 \\ 1 & \cot a & -l_0 \\ 1 & -\cot a & l_0 \end{bmatrix}, l_0 = r_x \cot a + r_y. \tag{15}$$

Having formulated the mathematical model, here, we use an example to illustrate how to determine the association graph according to the mathematical model. Considering (7)–(9), we can get the following derivations:

$$a_x = \sum_{i=1}^{4} (-1)^{i+1} \mathrm{sig}(\theta_i) T_i r_{\mathrm{motor}} \sin a \cos a / M \tag{16}$$

$$a_y = \sum_{i=1}^{4} \mathrm{sig}(\theta_i) T_i r_{\mathrm{motor}} (\sin a)^2 / M. \tag{17}$$

From (16) and (17), it can be concluded that the variables a_x and a_y are affected by four motor torque variables T_1, T_2, T_3, and T_4. Therefore, each motor torque variable has two edges connected to variables a_x and a_y, respectively, which means T_1, T_2, T_3, and T_4 are neighbors of a_x and a_y. The rest relationships can be modeled in the same manner, and the robot data association graph is constructed based on (7)–(15), as shown in Fig. 2. In the constructed robot data association graph, the self-connected edge is added for each sensor variable, but these self-connected edges along with the edges between variables v_1, v_2, v_3, and v_4 are drawn separately in Fig. 2 to make the figure more concise. Finally, the adjacent matrix A is generated based on the robot data association graph. If node j has an edge connected to node i, then the corresponding element of the adjacent matrix $A[i, j]$ is set to 1. Otherwise, $A[i, j]$ is set to 0, and the obtained adjacent matrix is shown in Fig. 3. From Fig. 3, it can be seen that sensor variables a_z, ω_x, and ω_y have no relationship with other sensor variables. However, these three variables contain the vibration characteristics of the wheeled robot, so they are also included in the constructed robot data association graph.

	v_1	v_2	v_3	v_4	E_1	E_2	E_3	E_4	T_1	T_2	T_3	T_4	a_x	a_y	a_z	ω_x	ω_y	ω_z
v_1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	1
v_2	1	1	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	1
v_3	1	1	1	1	0	0	1	0	0	0	0	0	0	0	0	0	0	1
v_4	1	1	1	1	0	0	0	1	0	0	0	0	0	0	0	0	0	1
E_1	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
E_2	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
E_3	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
E_4	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0
T_1	0	0	0	0	0	0	0	0	1	0	0	0	1	1	0	0	0	1
T_2	0	0	0	0	0	0	0	0	0	1	0	0	1	1	0	0	0	1
T_3	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	1
T_4	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0	0	0	1
a_x	0	0	0	0	0	0	0	0	1	1	1	1	1	0	0	0	0	0
a_y	0	0	0	0	0	0	0	0	1	1	1	1	0	1	0	0	0	0
a_z	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
ω_x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
ω_y	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
ω_z	1	1	1	1	0	0	0	0	1	1	1	1	0	0	0	0	0	1

Fig. 3. Adjacency matrix of the robot data association graph.

B. Spatial-Temporal Difference Graph Convolutional Network (STDGCN)

This subsection gives a detailed description of each component of the STDGCN. The core idea of the STDGCN is summarized as follows:

1) employing a difference layer for feature enhancement of graph nodes;
2) using the spatial-temporal graph convolution modules to capture spatial-temporal correlation features.

1) Difference Layer: After constructing the robot data association graph, we convert the sensor measurements of the wheeled robot into spatial-temporal graphs. Each node contains a monitoring value of the sensor variable in a moment. Since the graph nodes only contain the numerical features, we expect to extend the node feature by providing helpful extra information. To realize this objective, a difference layer is proposed for feature enhancement in this work. The designed difference layer makes use of localized difference properties by calculating the order 0 to D backward difference features. Then, the obtained multiorder backward difference features are taken as the new features of graph nodes.

Fig. 4 shows the calculation process of the difference layer, where $x^{(t)}$ denotes the tth time slice sensor measurements,

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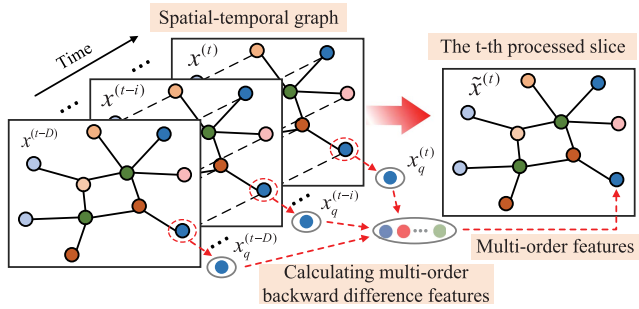

Fig. 4. Calculation process of the difference layer.

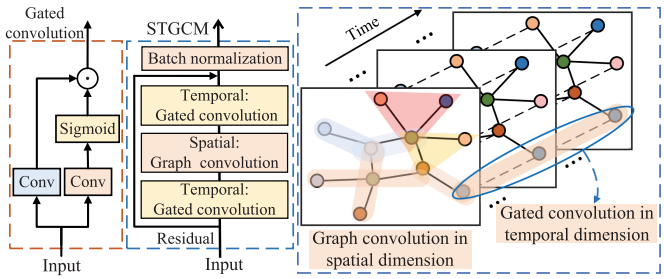

Fig. 5. Spatial-temporal graph convolutional module.

and these colorful nodes in $x^{(t)}$ represent different sensors' measurements at tth time slice. For the qth node in the t-th slice $x^{(t)}$ of a spatial-temporal graph, the node feature is $x_q^{(t)}$, and the computation of new node features by the difference layer can be defined as

$$\begin{cases} \Delta_0 x_q^{(t)} = x_q^{(t)}, \Delta_1 x_q^{(t)} = x_q^{(t)} - x_q^{(t-1)} \\ \Delta_2 x_q^{(t)} = \Delta_1 x_q^{(t)} - \Delta_1 x_q^{(t-1)}, \ldots \\ \Delta_D x_q^{(t)} = \Delta_{D-1} x_q^{(t)} - \Delta_{D-1} x_q^{(t-1)} \\ \tilde{x}_q^{(t)} = \mathop{\|}_{i=0}^{D} \Delta_i x_q^{(t)} \end{cases} \tag{18}$$

where $\Delta_D x_q^{(t)}$ denotes the calculated order D backward difference feature, $\|$ indicates the concatenate operation, and $\tilde{x}_q^{(t)}$ is the generated new node features. Note that (18) will meet the situation of $x_q^{(j\leq 0)}$ when $t \leq D$. To make the feature dimensions of nodes the same, we set $x_q^{(j\leq 0)}$ equals to 0. The generated multiorder backward difference features contain the sensor measurement change characteristics. For example, the order 1 and 2 backward difference features of sensor measurements v have the wheel acceleration and jerkiness characteristics. By taking advantage of these additional useful characteristics, the DL model is able to mine potential fault information.

2) Spatial-Temporal Graph Convolutional Module (STGCM): In wheeled robots, the multisensor measurements can naturally form the spatial-temporal graphs, and the fault information is included in the historical features of the nodes and their neighbors. Only modeling the temporal or spatial dependencies will lead to an unstable performance, because both temporal and spatial dependencies are crucial for fault diagnosis. Therefore, to capture fault features of both spatial and temporal domain, an STGCM is constructed as shown in Fig. 5. The STGCM consists of a graph convolutional layer, two gated convolutional layers, a residual architecture, and a batch normalization layer. The graph convolutional layer is utilized to mine spatial dependencies from neighboring nodes, and the stacked gated convolutional layer updates the features of a node by merging the corresponding node features at the continuous time slices. Each gated convolutional layer contains a gated linear unit [26], which is realized by two conventional 1-D convolutional layers, as presented in Fig. 5. The mathematical model of the gated convolutional layer can be defined as

$$y_{\text{gated}} = (K_1 * x_{\text{gated}} + b_1) \odot \sigma(K_2 * x_{\text{gated}} + b_2) \tag{19}$$

where K_1 and K_2 denote the convolutional kernels, b_1 and b_2 are the biases, and σ is the sigmoid function. x_{gated} and y_{gated} are the input and output of the gated convolutional layer. Note that the sizes of K_1 and K_2 are the same. The purpose of adding the residual architecture in STGCM is to let more information flow through. Moreover, the batch normalization layer (BN) is implemented to normalize the updated features to a fixed distribution, which is effective for reducing internal covariant shift.

C. Framework of the Proposed Method

The overall framework of the proposed STDGCN is illustrated in Fig. 6(a). Our STDGCN consists of a difference layer and two STGCMs. The difference layer calculates the backward difference features for feature enhancement, and the stacked STGCMs capture the localized correlation features in both spatial and temporal dimensions. After the STGCM, a temporal gated convolution is connected to compress the length of the node features. Besides, the global average pooling layer (GAP) is appended to aggregate features among the temporal dimension. Finally, all the features of graph nodes are concatenated and fed into the fully connected layer (FC) to perform fault classification.

The flowchart of the proposed fault diagnosis method for wheeled robot is shown in Fig. 6(b). We first build the robot data association graph according to the established robot mathematical model. Then, through converting robot sensor measurements on different time slices to graphs, the spatial-temporal graph dataset is generated. After that, we train the STDGCN using the training dataset and utilize the backpropagation algorithm to update its parameters. Finally, fault diagnosis is conducted using the trained STDGCN by inputting the testing dataset.

IV. Experiments

In this section, we will perform detailed ablation studies to verify the effectiveness of the proposed difference layer and the designed robot data association graph. Then, we compare the STDGCN with several existing fault diagnosis methods.

A. Dataset Description

The Mecanum wheeled mobile robot, developed by our research group, is taken as the experimental platform. This robot is driven by four brushless direct current motors and equipped with several kinds of sensors, including four encoders, an inertial measurement unit and so on. We mainly studies the diagnosis of five common kinds of faults as shown in Fig. 7, and Table III

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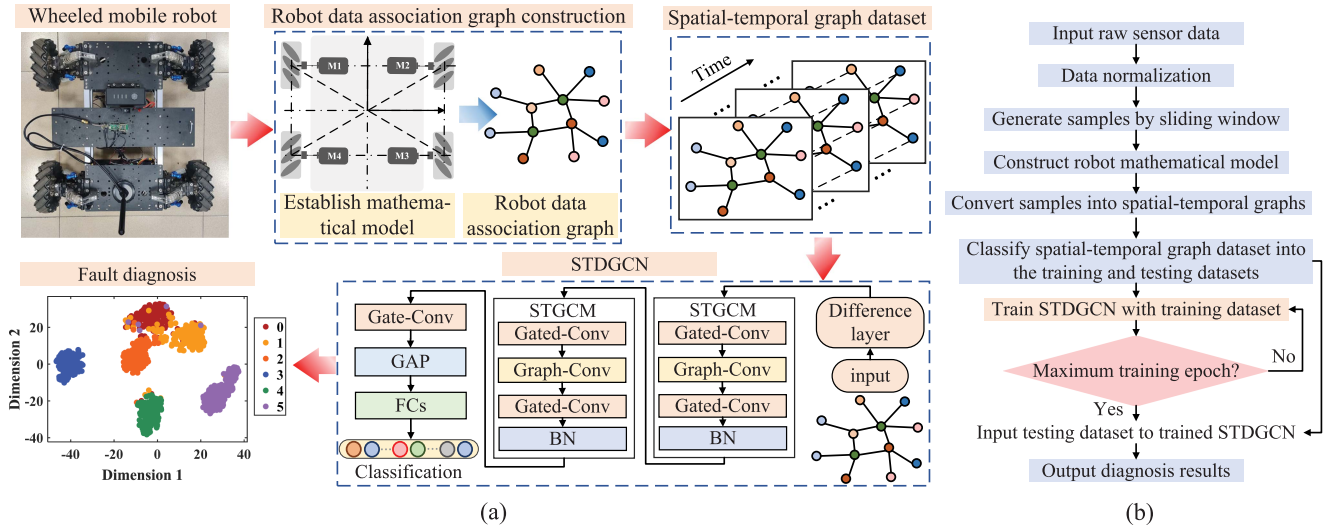

Fig. 6. Framework of the STDGCN and the corresponding flowchart for fault diagnosis of wheeled robot. (a) Framework of the STDGCN. (b) Flowchart.

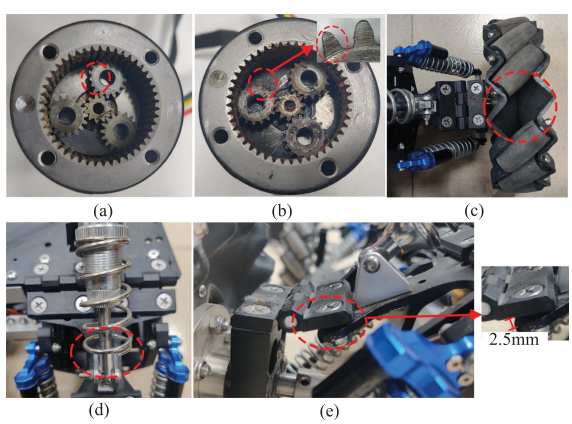

Fig. 7. Health conditions of the Mecanum wheeled mobile robot. (a) Motor gearbox tooth broken. (b) Motor gearbox tooth surface wear. (c) Wheel losing roller. (d) Shock absorber spring fail. (e) Shock absorber loosening.

TABLE III
FAULT POSITION AND MODE OF MECANUM WHEELED MOBILE ROBOT

Tag	Fault position	Fault mode
S0	Normal	-
S1	Right front motor gearbox	Tooth broken
S2	Right front motor gearbox	Tooth surface wear
S3	Right front wheel	Losing roller
S4	Right front shock absorber	Spring fail
S5	Right front shock absorber	Loosening

records the tag, fault position, and the corresponding fault mode. The Mecanum wheeled mobile robot runs under different health conditions to collect fault diagnosis dataset, and it is controlled to move in a straight line in a long corridor. The speed of the wheeled robot is set to 0.5 m/s, which is a common setting in an indoor environment. Multisensor data are collected with a sampling frequency of 100 Hz. Then, to eliminate the influence of data dimension and value range, we normalize the raw sensor data to the range of [0, 1] using the maximum–minimum normalization method. Besides, we employ the sliding window to segment the normalized sensor data into multiple samples by setting both step size and window length to 100. We randomly select 70% of samples as the training set, and the other 30% samples are taken as the testing set. It should be mentioned that the purpose of nonoverlapping sampling is to make sure that the testing set does not leak. Finally, a total of 4011 samples is generated, and the dimension of each sample is $x \in \mathbb{R}^{18\times 100}$.

TABLE IV
DETAILED STRUCTURE AND COMPLEXITY OF STDGCN

STDGCN		Output	FLOPs
Input		18×100×1	0
Difference Layer		18×100×5	7200
STGCM1	Gated convolutional layer_1	18×100×32	3686400
	Graph convolutional layer_1	18×100×32	7225600
	Gated convolutional layer_2	18×100×32	22348800
	Batch normalization layer_1	18×100×32	115200
STGCM2	Gated convolutional layer_3	18×100×64	44697600
	Graph convolutional layer_2	18×100×64	28787200
	Gated convolutional layer_4	18×100×64	88934400
	Batch normalization layer_2	18×100×64	230400
Gated convolutional layer_5		18×100×16	22233600
GAP layer		18×16	0
Fully connected layer_1		256	73728
Fully connected layer_2		6	1536

Considering the computing efficiency and diagnosis performance, the proposed STDGCN model consists of two STGCMs. The first STGCM contains 32 convolutional kernels along the temporal dimension, and 64 convolutional kernels are included in the second. In addition, the parameter D of the difference layer is set to 4, and the influence of this parameter on the model's performance will be discussed in the experiment. The detailed structure of STDGCN is illustrated in Table IV.

In this article, all the tested models are implemented by Pytorch 1.9.1, which is a well-known toolkit released by Facebook. Experiments are conducted on a workstation with an i9-10850 K CPU and two NVIDIA GeForce GTX3090 GPU. During the training phase, we employ the cross-entropy loss function and

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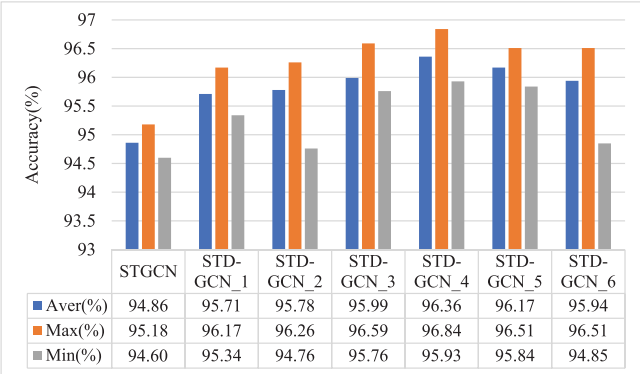

Fig. 8. Performance comparison of STGCN and the STDGCN models.

the Adam optimizer with the learning rate fixed at 0.001. We also set the batch size to 64 and each model is trained with 200 epochs. Five trials are performed in all experiments to reduce the influence of randomness on the results. Three performance indicators, i.e., maximum accuracy (Max), minimum accuracy (Min), and average accuracy (Aver), are taken as the evaluation metric to indicate the diagnosis performance of five trials.

B. Effectiveness of the Difference Layer

To investigate the effectiveness of the difference layer and the influence of the parameter D on the results, here, we construct six versions of STDGCN, i.e., STDGCN_1, STDGCN_2, . . ., STDGCN_6 and a degraded version of STDGCN, i.e., STGCN. Among these models, the STGCN model gets rid of the difference layer, and the last number in the name of these STDGCN models denotes the value of parameter D. Other settings of the constructed seven models keep the same. Experimental results are recorded in Fig. 8.

From Fig. 8, it can be seen that the difference layer can effectively improve the performance of the STDGCN. The Max and Aver of these STDGCN models are about 1% higher than that of the STGCN. Since multiorder backward difference features are used for feature enhancement, the STDGCN could learn better fault representations and yield higher diagnosis accuracy.

In addition, according to the comparison of these STDGCN models, we observe that as the value of parameter D increases from 1 to 4, the Aver of the STDGCN also increases. This reveals that the effects of multiorder backward difference features could be accumulated in the model, so as to continuously improving the diagnosis performance. Nevertheless, the Aver of STDGCN_5 and STDGCN_6 are slightly lower than that of the STDGCN_4, and the possible reason is that more difference features may distract the attention of the model. Therefore, the version STDGCN_4 is chosen as the final model.

C. Benefit of the Designed Robot Data Association Graph

This subsection conducts another ablation study to verify the benefit of the robot data association graph. Three scenarios are considered in the experiment, i.e., training using a self-adaptive association graph [18], training using a radius graph [27], and training using the robot data association graph. The self-adaptive association graph is parameterized module consists of $H \times H$ (H is the number of nodes) learnable parameters, and the value of H takes 18 here. The radius graph is realized by estimating the cosine similarity between nodes. If the calculated cosine similarity is greater than the threshold ϵ, there will be an edge between the two nodes. Same as [27], the value of threshold ϵ sets to 0 in this article. We build three models in the experiment: ASTDGCN, RSTDGCN, and STDGCN, where ASTDGCN is trained with the self-adaptive association graph and RSTDGCN is trained with the radius graph. Experimental results are summarized in Table V.

TABLE V
COMPARATIVE RESULTS OF ASTDGCN, RSTDGCN, AND STDGCN

Models	Max(%)	Min(%)	Aver(%)
ASTDGCN	95.43	93.85	94.85±0.57
RSTDGCN	95.10	94.44	94.83±0.27
STDGCN	96.84	95.93	96.36±0.29

From the comparison results, we could see that the diagnosis accuracies of ASTDGCN and RSTDGCN are almost the same, but the stability of the RSTDGCN is better than that of the ASTDGCN. The possible reason is that the use of self-adaptive association graphs increases the uncertainty in network training, thus ASTDGCN is unstable for spatial-temporal graph modeling. Moreover, training with the robot data association graph can boost the diagnosis accuracy, and the Aver of the STDGCN is about 1.50% higher than that of ASTDGCN and RSTDGCN. This is because the designed robot data association graph makes use of the prior knowledge, which can ensure the model explicitly mines the spatial correlations. These observations prove that the designed robot data association graph is beneficial for fault diagnosis.

D. Comparison With Existing Fault Diagnosis Methods

To evaluate the superiority of the proposed method, we compare the STDGCN with eight existing fault diagnosis methods, including recurrent neural network (RNN), gated recurrent unit (GRU), GCN, long short term memory network (LSTM) [28], residual network (Resnet), deep residual shrinkage network (DRSN) [29], multiattention 1-D convolutional neural network (MA1DCNN) [30], and the state-of-the-art spatial-temporal algorithm hierarchical attention graph convolutional network (HAGCN). The constructed RNN and GRU have two hidden layers, each hidden layer has 128 cells, and the length of time step and the dimension of input size are 100 and 18, respectively. The GCN implemented in the experiment has two graph convolutional layers and the Resnet adopts the Resnet-18 architecture. Moreover, the detailed setting of the models LSTM, DRSN, MA1DCNN, and HAGCN can be found in their articles. Experimental results are recorded in Table VI, and the best results are shown in bold. According to these results, several observations could be obtained.

First, the Max, Min, and Aver of the RNN are worse than those of other compared methods. Since the RNN has the problem of vanishing gradient and its structure is simple, it is difficult for RNN to handle long temporal sequence, resulting in a poor diagnosis performance.

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TABLE VI
COMPARISON RESULTS OF STDGCN AND OTHER METHODS

Models	Max(%)	Min(%)	Aver(%)
RNN	62.29	51.66	58.33±3.75
GRU	93.85	92.27	93.23±0.59
LSTM [28]	92.69	90.78	92.04±0.65
Resnet	86.37	85.63	86.06±0.30
DRSN [29]	84.05	78.57	80.83±2.09
MA1DCNN [30]	87.95	77.15	80.49±4.18
GCN	90.78	89.95	90.34±0.34
HAGCN	93.02	91.11	92.10±0.60
STDGCN	**96.84**	**95.93**	**96.36**±0.29

Second, three CNN-based models, namely Resnet, DRSN, and MA1DCNN, do not perform well on the wheeled robot fault diagnosis dataset as the Aver of these models are less than 90%. It could be ascribed that these CNN-based models are not suitable for wheeled robot data fusion. To be specific, in the experiments, measurements from 18 sensors are directly inputted into the CNN-based models. The conventional convolutional kernels generate new feature maps by aggregating the features of all channels, regardless of whether they are related or not. As a result, the important fault information in some channels would be affected by the irrelevant and redundant information in other channels, which has a negative effect on the model performance.

Third, GRU, LSTM, and GCN outperform the compared CNN-based methods. The Aver of these three methods reach 93.23%, 92.04%, and 90.34%, respectively. The possible explanation is that the way they extract features is better than CNN-based methods. GRU and LSTM adopt the gated mechanism to learn the temporal dynamics from the input, while the GCN aggregates adjacent node features based on the Laplacian matrix of association graph. However, these models only focus on spatial or temporal dependencies, thus they are suboptimal for wheeled robot fault diagnosis.

Finally, the Aver of the HAGCN is 92.10%, which is about 1.76% higher compared to that of the GCN. However, there is still room for improvement in the performance of the HAGCN, as the used regularized self-attention graph pooling layer causes the loss of useful fault information in some nodes. The proposed STDGCN achieves the best diagnosis result among all the tested methods. The Max, Min, and Aver of the STDGCN are 96.84%, 95.93%, and 96.36%, respectively, with a standard deviation of 0.29%. The direct reasons for the higher accuracy of the STDGCN are summarized as follows:

1) the proposed method makes use of the prior knowledge of the wheeled robot so that the STDGCN could stably exploit the spatial correlations of sensor measurements;
2) the STDGCN incorporates the difference layer, which enables it to catch more useful information;
3) the STGCM introduced in the STDGCN could model the spatial-temporal features along with the dynamic correlations.

The t-distributed stochastic neighbor embedding technique is adopted to visualize the learned feature representations at the last layer of four selected models, i.e., Resnet, GCN, LSTM, and STDGCN. Though mapping higher dimensional features into the 2-D, visualizations could intuitively show whether these feature representations are discriminative or not. The visualization results are drawn in Fig. 9.

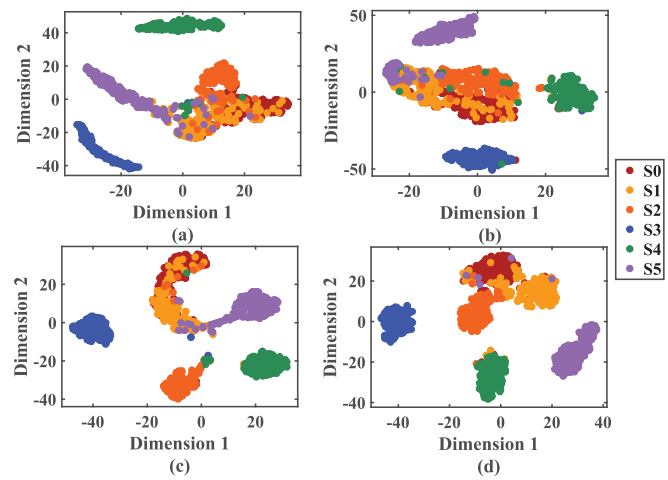

Fig. 9. Visualizations of the learned feature representations at the last layer in (a) Resnet, (b) GCN, (c) LSTM, and (d) STDGCN.

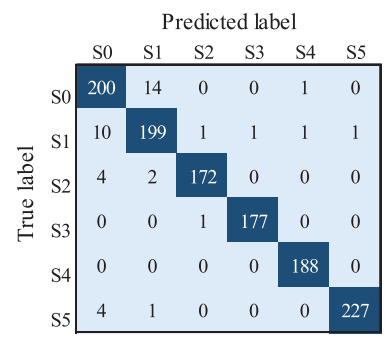

Fig. 10. Confusion matrix of the STDGCN.

From Fig. 9(a), it can be observed that the generated 2-D features of Resnet are highly mixed. Due to the fault information is mixed with irrelevant and redundant information, the learned feature representations in Resnet are not discriminative enough. From Figs. 9(b) and (c), it can be seen that the 2-D features of GCN and LSTM still suffer severe overlap. As GCN and LSTM ignore temporal or spatial dependencies, the extracted features cannot be completely separated. Fig. 9(d) indicates that the feature distributions of the STDGCN are relatively easier to distinguish and more concentrated, which means the STDGCN has a better feature extraction ability. These findings are consistent with the previous observations. Fig. 10 clearly shows the confusion matrix of the STDGCN (a randomly selected trial). The rows represent the true labels, while the columns denote the predicted labels. It can be concluded that the main errors come from the wrong classification between health conditions S0, S1, and S2. This is due to the low severity of gearbox faults, which causes the fault characteristics not obvious.

E. Time Complexity Analysis of the STDGCN

This subsection carries out the time complexity analysis of STDGCN by estimating the number of floating-point operations

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TABLE VII
AVERAGE TEST TIME OF STDGCN AND OTHER METHODS

Models	Test time (s)	Models	Test time (s)
RNN	7.30×10^{-3}	MA1DCNN [30]	3.38×10^{-2}
GRU	1.21×10^{-2}	GCN	2.37×10^{-3}
LSTM [28]	5.13×10^{-3}	HAGCN	4.49×10^{-2}
Resnet	7.23×10^{-2}	STDGCN	5.98×10^{-2}
DRSN [29]	2.06×10^{-2}	-	-

(FLOPs). The computational cost of the proposed STDGCN comes from the gated convolutional layer, the graph convolutional layer, the difference layer, the batch normalization layer, and the fully connected layer. As aforementioned, each gated convolutional layer contains two conventional convolution operations. The time complexity of the convolutional layer (Conv) is given by Molchanov et al. [31], which can be expressed as

$$TC_{\text{Conv}} = 2H_{\text{in}}W_{\text{in}}(C_{\text{in}}K_{\text{conv}} + 1)C_{\text{out}} \tag{20}$$

where H_{in}, W_{in}, and C_{in} represent the height, width, and channels of the input feature map, respectively, K_{conv} denotes the kernel size, and C_{out} indicates the output channels.

For the graph convolutional layer (GConv), the time complexity can be defined as [32]

$$TC_{\text{GConv}} = LN_{\text{node}}D_{\text{node}} + LN_{\text{edge}}D_{\text{node}}^2 \tag{21}$$

where L is the number of layers, N_{node} and N_{edge} are the number of nodes and edges in the graph, and D_{node} represents the dimension of node features. In addition, the time complexity of the difference layer (DL), the batch normalization layer (BN), and the fully connected layer (FC) can be defined as

$$TC_{\text{DL}} = D_{\text{in}}D \tag{22}$$

$$TC_{\text{BN}} = 2D_{\text{in}} \tag{23}$$

$$TC_{\text{FC}} = D_{\text{in}}D_{\text{out}} \tag{24}$$

where D is the parameter of the difference layer, which means calculating order 0 to D backward difference features. D_{in} and D_{out} denote the dimension of the input and output feature maps, respectively.

According to Table IV, we can compute the total FLOPs of STDGCN using the following equation:

$$TC_{\text{STDGCN}} = \sum_{i=1}^{10} TC_{\text{Conv}} + \sum_{i=1}^{2} TC_{\text{GConv}} + \sum_{i=1}^{2} TC_{\text{BN}} + \sum_{i=1}^{2} TC_{\text{FC}} + TC_{\text{DL}}. \tag{25}$$

It is worth mentioning that the size of all convolutional kernels is set to 3, and the values of N_{node} and N_{edge} are 18 and 70, respectively. The calculated total time complexity of the STDGCN is 218341664 FLOPs, and the detailed time complexity of STDGCN is listed in Table IV.

Table VII lists the average test time of all methods in five trials to compare the computational efficiency of the STDGCN with other methods. Note that test time is the time consumption of predicting labels of the whole testing set (1204 samples). According to Table VII, it can be calculated that the test time of all models for a single sample is less than 1×10^{-4} s, which means all these trained models satisfy the needs of online fast fault diagnosis of wheeled robot. Comparing the results in Table VII, it can be observed that the test time of RNN, GRU, LSTM, and GCN is shorter than that of other models, which is due to their simple structures. In addition, the test time of DRSN, MA1DCNN, HAGCN, and STDGCN has no significant difference, whereas the STDGCN is slightly more time consuming than the other three methods. This is because the STDGCN captures the spatial-temporal correlations simultaneously, which is beneficial for fault diagnosis but costs more time. Moreover, the test time of Resnet is the highest because it has more layers and convolutional kernels than other methods.

V. CONCLUSION

This article proposed a novel wheeled robot fault diagnosis method that integrated prior knowledge and the spatial-temporal difference graph convolutional network. The main advantages of our proposed method are summarized as follows:

1) the prior knowledge of the robot system was adopted to construct the association graph so that the GCN-based model can accurately capture the spatial dependencies among sensor measurements;
2) a difference layer was proposed to calculate multiorder backward features for feature enhancement of graph nodes;
3) the spatial-temporal convolutional module was introduced to simultaneously model the spatial-temporal dependencies.

Based on the Mecanum wheeled mobile robot dataset, ablation studies and comparative experiments were carried out. Experimental results revealed that the proposed difference layer and the constructed robot data association graph were beneficial for fault classification, and the developed STDGCN obtained the state-of-the-art performance.

The proposed method can be applied to four Mecanum-wheeled mobile robots. For other types of mobile robots, such as three-wheeled or two-wheeled mobile robots, we recommended establishing the simplified robot kinematic and dynamic model, building the robot data graph representation, and then, applying the proposed STDGCN for fault diagnosis. Moreover, the idea of making use of prior knowledge and the proposed STDGCN model can be used for reference for other fault diagnosis cases.

The limitations and future work are summarized as follows: this article had made the assumption that the number of samples in each category was almost the same, which was also the limitation of this study. In practical application, the amount of normal data were far more than that of faulty data, which caused the sample imbalance problem. In future work, we will expand the wheeled robot fault diagnosis dataset and investigate how to alleviate the influence of sample imbalance. Generally, for the fault diagnosis with imbalanced sample data, the data augmentation based methods (such as generative adversarial networks, data oversampling, etc.) and the feature adaptation based methods (such as joint distribution adaptation, transfer

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component analysis, etc.) are effective tools to solve this kind of problem. We will study the most effective method to solve the sample imbalance problem for wheeled robot fault diagnosis through experiments.

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